

Measuring Broadband Performance in South Africa

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ABSTRACT

Broadband penetration is increasing in South Africa, particularly on mobile devices, but little is known about the performance of fixed or mobile broadband in the country. This lack of empirical data imposes significant limitations to innovation, because broadband performance metrics help users audit their connectivity costs and regulators to make informed decisions about policies and infrastructure investments. We present the results from a measurement study of both fixed and mobile broadband connections in South Africa. We use measurement software that we implemented on mobile phones and home routers and adapted to address challenges that are unique to the developing world. Our findings suggest that consumers in South Africa are not getting advertised speeds, that mobile broadband generally has higher throughput than fixed broadband, and that interconnection between ISPs (or lack thereof) plays a significant role in determining the reliability and performance that users ultimately receive. Often, high latencies to destinations introduce significant performance bottlenecks, suggesting that, in addition to investments in higher throughput links, effort should be devoted to improving interconnection between ISPs and locating content closer to users. Our study, along with our open source technical platform, embodies an archetypal method for monitoring broadband performance in developing countries.

Categories and Subject Descriptors

C.2.3 [Computer-Communication Networks]: Network Operations—Network Management; K.4.1 [Computers and Society]: Public Policy Issues – *regulation*.

General Terms

Management, Measurement, Performance

Keywords

Broadband performance, South Africa, policy, quality of service, consumer protection, broadband measurement platform

1. INTRODUCTION

Despite increasing broadband penetration in South Africa [16,28] particularly on mobile devices [30], there is little empirical data about the country's broadband performance. This lack of data is troubling: aside from cost [10], performance affects broadband adoption and use [8,35] which, in turn, is associated with progress and development [34]. Monitoring broadband performance is necessary to ensure that consumers are receiving the service level that they paid for, that their connections are reliable, and that regulators can make informed decisions [13,23,24]. Yet, to our

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knowledge, no systematic study of fixed and mobile broadband performance has been conducted in South Africa (or anywhere in Africa, for that matter). Moreover, unlike more developed countries, broadband access in South Africa is more costly, the mobile infrastructure is more extensive than the fixed line infrastructure, and there are more mobile than fixed-line users [17]. These conditions call for a new approach for measuring broadband performance that is tailored to the constraints of a developing country.

This paper addresses this gap in our knowledge. We adapted our open-source broadband measurement platform—custom OpenWrt-based home router firmware and an Android mobile speed test application—to the conditions in South Africa. Next, we gathered measurements about the quality of fixed and mobile broadband connectivity in the country, where mobile refers to Internet access via a mobile phone or using a 3G “dongle” (*i.e.*, a 3G USB modem). We studied the following questions:

- Does the performance of each user's fixed or mobile broadband connection achieve the performance advertised by the Internet service provider (ISP)?
- How does fixed-line broadband performance compare to mobile broadband performance?
- What performance do users in South Africa experience to commonly accessed Web sites? When performance suffers, what are the causes?

To answer these questions, we deployed our custom home network routers in 16 testing sites in all nine provinces (separately governed regions) across 15 service providers in South Africa. We then collected regular throughput and latency data for approximately three months in 2013. We also collected mobile performance data from several hundred mobile phones using our mobile speed test application, and mobile performance measurements from two 3G “dongles” over several weeks. We augment our datasets with data collected from fixed and mobile devices using a custom `speedtest.net` tool hosted by the local website MyBroadband [22] during the same time period.

We make three contributions: First, we offer the first publicly available measurement data for both fixed and mobile broadband performance in South Africa (all of our measurement data is publicly available). Second, we describe the challenges that we faced when performing broadband performance measurements in a developing country and how we addressed these challenges. Finally, we use examples from our data to illustrate factors in local infrastructure that introduce significant performance bottlenecks in South Africa (and more generally across Africa).

Many of our findings are surprising and significant. First, many users appear to receive broadband speeds that are less than what they pay for. This stands in contrast to similar studies from the United States (US) and United Kingdom (UK), both of which found that ISPs generally deliver their advertised rates [24,31]. Second, mobile providers typically offer faster download speeds to users than fixed-line ISPs do, although mobile performance tends to be significantly more variable than fixed-line performance. Third, and perhaps most importantly, the throughput

Data Set	Collection Period	Measurements
BISmark	Feb 2013–April 2013	Upload and download throughput Latency to local and international servers TCP connect times for popular websites
MyBroadband	11 Feb 2013–9 April 2013	Upload and download throughput Latency to local and international servers
MySpeedTest	22 Nov 2012–31 April 2013	Latency to local and international servers

Table 1. Summary of data sets used in analysis.

that an ISP provides is only part of the story—equally important is the *latency* to Web sites and services that users actually visit. Both the physical distance to servers for popular sites (*e.g.*, Facebook) and the ISP-level connectivity (or lack thereof) between users and these servers can introduce latencies of several hundred milliseconds. These delays are far too high to provide consistently good performance, even if throughput were to improve.

Our results demonstrate that there is a distinct need to provide consistent, continual, and open measurements of broadband performance for ISPs in developing regions. In addition to

ISP	# Users
Afrihost	1
I-AXXESS	1
MWeb	1
Telkom	9
WebAfrica	3
V1	1
8ta (3G “dongle”)	1
MTN (3G “dongle”)	1
Total	18

Table 2. ISPs for DSL line and BISmark users.

increasing accountability, the data we collect can help policymakers make arguments concerning regulatory or financial incentives for companies that provide popular services to deploy infrastructure closer to users in South Africa. Although the results in this study are specific to South Africa, we expect that many of the general findings may hold in similar developing countries. We are expanding our study to other African countries in future work.

2. BACKGROUND

We motivate the need to measure broadband performance and discuss the context of broadband in South Africa, including metrics and methods for measuring broadband performance.

2.1 Why Measure Broadband Performance?

Broadband penetration in developing countries, particularly in sub-Saharan Africa (SSA), remains sparse, although mobile broadband (both 3G and cellular connections) far outpaces fixed-line broadband deployments [16]. Where connectivity exists in SSA, many applications designed for social and economic development take advantage of this connectivity [14]. Previous work suggests that slow speeds and limited bandwidth in SSA constrains the user experience by forcing users to be more deliberate in their browsing behaviors [26,35]. Others have tried to address these problems through improved system design in similar developing countries that experience poor broadband performance [9]. Unfortunately, in many cases, the quality of the connectivity is unknown. Canadi *et al.* have studied broadband

performance [7], but they do not focus on either SSA or South Africa, nor do they perform longitudinal measurements or measurements of mobile broadband. Given the extent of mobile broadband access on the continent (93 million mobile broadband subscribers versus 2 million fixed broadband subscribers in 2013 [17]), the lack of research on mobile broadband quality of service in particular must be addressed. Towards this goal, we provide the first systematic study of both mobile and fixed-line broadband performance in South Africa.

2.2 Broadband Connectivity in South Africa

Broadband adoption in South Africa has been slow but growing since 2003 [18,19]. Most fixed-line access is via Asymmetric Digital Subscriber Line (ADSL) connections. Internet access via a “dongle” or 3G USB modem has also become popular because of the mobility of the connection and the relatively low cost of access [30]. These dongles typically involve lower setup costs and less maintenance as compared to fixed lines. For these reasons,

Mobile ISP	# Measurements
Vodacom	93,836
MTN	73,957
Cell C	44,419
Total:	212,212

Table 3. Number of latency measurements collected from each ISP using MySpeedTest.

many users opt for a 3G “dongle” as their primary Internet connection at home. In contrast to fixed-line access, which is seeing slow but steady growth, mobile broadband access is rapidly growing in the country, as it is across Africa [17].

Cost still remains a limiting factor for widespread broadband adoption and use in the South Africa [10]. For instance, uncapped Internet plans are still prohibitively expensive, and many users still use capped plans as low as 1 GB per month for fixed-line access or 10 MB for mobile access. Speeds vary from as low as 1 Mbps to 40 Mbps on Very-high-bitrate DSL (VDSL); the latter is restricted to certain areas, with each tier being more costly [30]. On mobile networks, Long Term Evolution (LTE) or 4G speeds can reach 60 Mbps but in practice is closer to 15 Mbps.

In SSA, South Africa is relatively progressive in terms of regulating broadband because it has higher rates of broadband penetration [16]. The government has been working on a national broadband policy for several years [12], and a draft national broadband policy was released in April 2013 [27]. As of yet, the country’s policy draft does not specifically address how (or even whether) the quality of broadband connections will be regulated or monitored. Our research shows that such continued monitoring can inform regulators about actual broadband performance for crafting policies based on sound empirical measurements.

3. DATA COLLECTION

We first describe commonly used metrics and methods for measuring broadband performance and existing data sets on performance. We then describe our technical platform for measuring broadband performance and how we deployed it in South Africa, including our process for recruiting users. Finally, we describe the technical and logistical challenges associated with performing broadband measurements in a developing country context and how we addressed them.

3.1 Metrics and Measurement Methods

We now describe the metrics and methods that we use for measuring fixed and mobile broadband performance. We then describe existing data on broadband performance in South Africa.

3.1.1 Performance metrics

Common metrics for measuring broadband performance include upload and download throughput (“speed”), latency, jitter, and packet loss [5,13,24,31]. We explain each of these metrics below. The methods we describe are generally accepted broadband performance tests [32].

Upload throughput, usually reported in megabits per second (Mbps), indicates how fast a user can transfer data. Typical applications affected by upload throughput include bulk transfer (e.g., uploading videos or photos to a website).

Download throughput measures throughput in the reverse direction (from the destination to the user). Applications affected by download throughput include video streaming and file sharing. Throughput is often measured by measuring the time to transfer a file between two machines on the network.

Latency, usually reported in milliseconds (ms), is the time for data to travel from one endpoint to another. Latency affects real-time applications such as Voice over IP and gaming. Latency measurements are conducted using ping tests, a network function that sends a packet of information to a server and measures the time it takes to get a response.

Jitter measures the consistency of latency; it is also reported in milliseconds (ms). High jitter affects real-time applications such as gaming, voice, and video streaming.

Packet loss is reported as a percentage and refers to the number of packets that do not reach their destination. High packet loss degrades the performance of real-time streaming and interactive applications such as voice and video conferencing; it also affects any application requiring reliable connectivity, including web browsing and file transfers.

3.1.2 Measurement methods

For fixed-line broadband connections, *host-based* measurements are conducted from a user’s end device or “host”, such as a computer or mobile phone. A user might run a software application on a device by downloading an application [5], clicking on a web application, such as the popular and well-known speedtest.net created by OOKLA [25], or by measuring performance when a user accesses content from a particular known server, such as tests conducted by the popular content distribution network, Akamai [2]. Unfortunately, host-based measurements often require the user to initiate the test, and the results can also vary depending on factors inside the local environment, such as the wireless network, the user’s device, or multiple users using the same connection. These tests offer snapshots of performance, rather than in-depth data on one user over time [7]. Given that performance varies continuously, these

Mobile ISP	# Measurements
8ta (mobile)	9,073
Cell C (mobile)	2,553
iBurst (mobile)	2,861
MTN (mobile)	12,798
Vodacom (mobile)	8,662
Total:	35,917
Fixed ISP	# Measurements
Afrihost	7,364
Cybersmart	1,991
iBurst	149
Internet Solutions	6,496
MWeb	9,721
Neotel	188
Telkom	11,571
Vox	557
WebAfrica	614
Total	38,651
Mobile + Fixed Total	74,568

Table 4. Throughput and latency measurements collected from MyBroadband (each measurement instance has one set of throughput measurements and one latency measurement).

measurements do not completely capture the performance a user experiences over time [13,31]. There is also no way to ensure that users will perform regular measurements or to verify certain information that the user provides.

Router-based measurements use a router equipped with custom firmware to measure broadband performance. A user replaces their router or connects the custom router to their existing router at the site where broadband performance is to be measured such as in a private residence (see Figure 1). Because the router is always on and connected directly to the provider, these tests can offer continuous measurements and arguably yield a more accurate measurement by accounting for environmental factors [32]. Router-based measurements can be continuous, and no user attention is required after installation. The disadvantage—particularly in a developing country—is that routers are relatively expensive and require more effort to deploy and maintain.

Mobile broadband performance measurements are host-based and conducted via a web application through a mobile browser or a software application downloaded to the mobile phone [23]. In general, mobile measurements are more difficult to interpret because the factors affecting mobile performance are much different than on a fixed line [5]. For example, congestion in a particular area or time of day where multiple people are accessing the network in the same area (e.g., a soccer match) can affect the speeds that users observe.

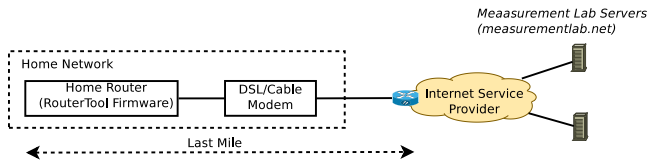


Figure 1. Illustration of router-based measurement. We install custom firmware on the user’s home router and measure throughput, latency, and jitter to globally distributed Google Measurement Lab servers

3.1.3 Existing measurements in South Africa

There is scant data on broadband performance in South Africa. Limited latency data exists for broadband performance for SSA in general. For instance, the RIPE Atlas project [29] uses custom probes to measure network reachability but they do not conduct speed tests and their coverage of Africa is low. PingER similarly reports latency measurements for certain African countries [11]. Aggregated host-based measurements on broadband performance for many countries in SSA do exist, including South Africa. For example, according to OOKLA’s NetIndex [25], South Africa ranks 126th for speeds in the world with an average speed of 3.49 Mbps for downloads and 138th in the world with an average of 1.33 Mbps for uploads. Akamai [3] similarly reports data for many African countries including South Africa (2.149 kbps average connection speed).

Although the OOKLA and Akamai datasets have thousands of data points, they do not offer in depth insights into users’ networks, service plans, and performance problems, or how performance varies over time. They provide less insight into the performance per country, and the detailed measurement data is not public. Although we have fewer measurement vantage points than the Akamai or OOKLA datasets, our study provides a more comprehensive view of South African broadband performance on both mobile and fixed connections because we use both host-based and router-based measurements, and we perform continual measurements over time from the same locations. Our analysis also goes beyond Canadi *et al.*’s [7] examination of OOKLA data for several African countries, which did not include South Africa.

3.2 Data Collection: Deployment and Tools

Our study and data collection is ongoing; this paper reports on data between February 2013 and May 2013 in South Africa, although in some cases we use additional data (see Table 1). We collected data on broadband performance on fixed-line and mobile connections using both host-based and router-based measurements from an open-source measurement platform that we implemented and adapted for the South African environment. Where required, we recruited users through a popular local website, MyBroadband [22], local email lists, social media sites such as Facebook.com, Twitter.com, word of mouth, and through our local contact network and organization website. We use two tools to measure performance that users experience: BISmark¹ and MySpeedTest¹. Using these tools, we have been collecting data around the world in 30 countries since 2010. Our previously published work provides more technical details about these tools and the tests conducted [31,32]. In this paper, we focus on how we have adapted these tools to operate in a developing country, where constraints are different. To ensure that we measure bottlenecks introduced by the network, rather than our own infrastructure, we

conduct performance measurements to well-provisioned servers hosted by Measurement Lab (<http://measurementlab.net/>).

BISmark: To measure fixed-line broadband performance, we recruited 16 testing sites to use our BISmark router (see Table 2). BISmark is custom firmware that we built on OpenWrt that performs router-based performance measurements and sends data to a collection server several times a day. We deployed BISmark routers at 14 residential sites, one hospice, and one office. We specifically recruited a set of users that comprised at least one router in each of the nine provinces in South Africa. In total, we had seven users in the Western Cape, two in Gauteng, and one in each of the seven remaining provinces.

Each site was provided with a BISmark router. At one location in Cape Town, Western Cape, we also plugged two 3G “dongles” into two dedicated routers in addition to the router measuring the fixed line at that same location. We asked all users to complete a demographic questionnaire to provide information about household (or office) members, details on Internet service plan, ISP, and their perceptions of their Internet speed. We performed continuous measurements for the metrics in Table 1. We also measured packet loss and jitter, and TCP connect times to the top 100 websites in South Africa as ranked by Alexa [4] in April 2013. All throughput measurements during the study were performed to a Google Measurement Lab server in Johannesburg, South Africa. We also performed latency measurements to servers in Africa, Asia, America, Australia, and Europe.

MySpeedTest: We recruited mobile broadband users to download and install MySpeedTest, our mobile speed test application, on their Android phones (see Table 3). MySpeedTest allows users to perform throughput measurements on demand; it also collects latency and metadata automatically, and users are provided with summaries of their data performance in a set of graphs. We offered an Android phone prize for the user who conducted the most throughput tests during the study period. Our user set was limited to Android users because the software application currently only runs on these phones. We collected continuous latency measurements to the same set of servers as with BISmark, as well as metadata about the users’ phones and ISPs. Because the throughput measurements we collected during this period were conducted to an international server (which may ultimately reflect a throughput bottleneck that is not local to the country), we do not discuss them; we focus instead on latency for this dataset.

MyBroadband: In addition to our own deployments, we gained generous access to data collected using a custom speedtest.net server run by the MyBroadband website in South Africa (see Table 4). Users run these host-based speed tests from their website, selecting their service plan and whether to run the test to a local (Johannesburg) or international (London) server. These measurements also have a timestamp, IP address, ISP name, and latency in milliseconds to the server the user selects.

3.3 Challenges in a Developing Country

We encountered several challenges in measuring broadband performance in a developing country that required us to modify our existing measurement approach for local conditions. First, we had to reduce the frequency of measurements in both BISmark and MySpeedTest because of the high cost of sending data traffic in South Africa and the presence of data caps, which limit the total amount of traffic a user can download in a month. We could not assume that all participants would be able to download unlimited amounts of data. For example, we had five home users on capped plans, with caps ranging from 1–130 GB per month.

¹ We have implemented “BISmark” and “MySpeedTest”; these tools are publicly available. The “MyBroadband” data was collected from a tool that we did not implement.

Determining the appropriate frequency for measurements presents a tradeoff: higher frequency provides better data about time-varying properties of a user's Internet connection but also risks exhausting the user's data quota for a particular month or incurring significant data costs. We opted for speed measurements every eight hours on fixed lines and on demand for mobile devices. Because latency measurements require less data, we did not have to adjust the frequency of our latency metrics.

Second, recruiting participants was difficult both because many users had privacy concerns (even though the tool does not collect any user data, some people had difficulty understanding or believing this was the case), and because there are not many fixed-line broadband users in the more sparsely populated areas of South Africa. Other developing countries may have less fixed broadband penetration in rural areas making measurements harder to gather. In the future, we believe that this barrier to achieving more participants on fixed lines could be overcome if the ISPs who are deploying customer premises equipment (CPE) such as modems and routers would automatically deploy routers with custom measurement firmware (such as BISmark) when a customer subscribes for their service. For mobile recruiting, we had to find innovative ways to encourage adoption of MySpeedTest. If the target areas for measuring mobile broadband performance are in lower income or more sparsely populated areas, creative ways for incentivizing use of the app is needed. Giving away free data bundles is one way that could encourage use; we opted for awarding a prize instead.

Third, we had to ensure that our measurements would not disrupt users' Internet access. The router-based approach that we discussed earlier helps obtain performance metrics over time, since it is always connected to the Internet, as long as the device does not disrupt the user's Internet connection. We carefully designed BISmark to conduct throughput measurements at a frequency and duration that did not interfere with users' Internet performance: The measurements are brief (typically less than 15 seconds) and infrequent. MySpeedTest only performs throughput tests on demand.

Finally, keeping users engaged in the study is both difficult and important. If users are not engaged, they may simply discard or unplug the equipment, or even blame their poor Internet connectivity on our measurement tools (and stop using it). Engagement is crucial in cases where the measurement equipment must be managed physically, such as for the router-based measurements, or when we needed to gather additional information from the user (e.g., about their Internet speed). In many cases, one way to keep users engaged in the study was to present them with meaningful data about their Internet connection. For example, our BISmark uploads the performance data to Amazon Web Services, where a user can download and analyze his or her own data; we have also developed a public Web portal that allows the user to easily examine their ISP's performance with graphs and visualizations. Similarly, MySpeedTest presents users with a summary of their performance.

4. RESULTS

We now highlight the results from our study using data from the BISmark, MySpeedTest, and MyBroadband datasets. We first study whether fixed-line and mobile ISPs provide the level of service that they advertise to their users. We then explore other factors that can affect performance. Although our results may not reflect general performance trends (e.g., one cannot necessarily

conclude that one ISP provides better performance than another), we highlight trends where they do exist, and, where possible, correlate our findings across all three data sets. Our findings may not be indicative of performance in low-income townships in urban areas since we did not measure performance in these areas. Fixed-Line & Mobile Access Performance

We examine both fixed-line and mobile performance for 15 ISPs across South Africa. We first explore how measured downstream throughput compares to advertised rates. We also examine the network latencies of various ISPs.

4.1.1 Does performance match advertised rates?

We studied whether the throughput that different ISPs achieve matches the rate that these ISPs advertise to their customers. The advertised rates at the various fixed line testing sites were 512 kbps, 1 Mbps, 2 Mbps, 4 Mbps, and 4-10 Mbps. We introduce the notion of *normalized throughput*, which is the throughput that we measure for the fixed line connection (as described in Section 3.2) divided by the throughput that the ISP advertises for the plan that it sells to a user. Figure 2 shows the normalized throughputs for each of the ISPs in the BISmark data set (the caption describes how to interpret the box plot). The results show that, in most cases, users in our study do not achieve their advertised rates. Occasionally, some users see throughput that exceeds the advertised rate, as shown by the whiskers.

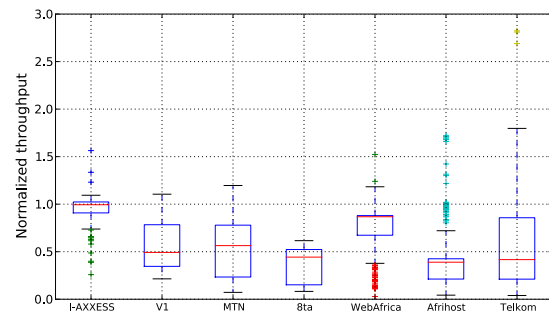


Figure 2. The throughput for each ISP normalized by the rate that the ISP advertised for each service plan (1.0 means that the throughput meets the advertised rate). The box shows the inter-quartile range of the normalized throughput; the line in the box depicts the median value; and the whiskers show 10th and 90th percentiles. *6 of the 7 ISPs examined fall far short of meeting their advertised rates. (BISmark)*

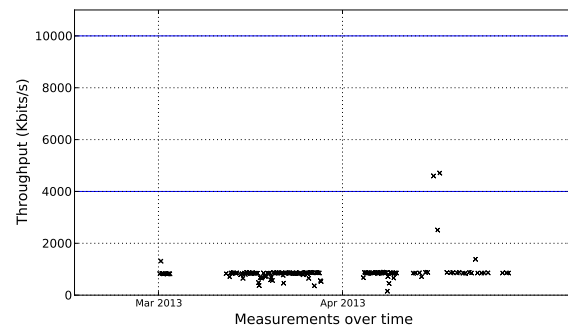


Figure 3. A timeline of throughput measurements from a Telkom user in Cape Town, South Africa. The advertised downstream throughput for this ISP is 4–10 Mbps; unfortunately, the user only sees about 1 Mbps most of the time. (BISmark)

Upon further investigation, we witnessed several egregious cases where a user was receiving throughput speeds that were significantly less than advertised rates. For example, Figure 3

shows a timeline of throughput measurements for approximately two months for a Telkom user with a service plan that advertised rates of 4–10 Mbps. The figure clearly shows that the downstream throughput of the user’s service plan does not meet the advertised rate. This example shows an extreme case, but the aggregate results also show the same trend.

This result makes sense, given that mobile infrastructure is more developed than fixed-line infrastructure. Still, the magnitude of the performance difference contrasts starkly with the findings from similar studies in developed regions [7,31].

Latencies. We also studied the distribution of latencies to a server in Johannesburg for different ISPs in our BISmark data set. Figure 4 shows this distribution, with a log scale on the y-axis. The results show that latencies are incredibly variable and that some ISPs have much more variable latency than others.

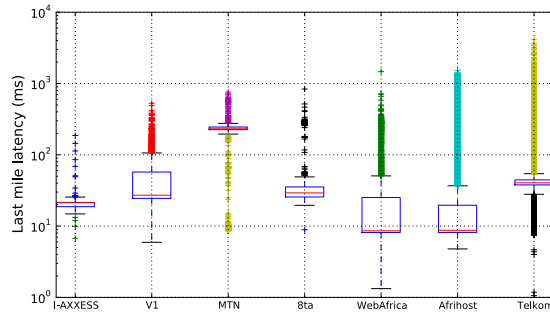


Figure 4. The latencies for each ISP to a server in Johannesburg, South Africa. The boxes show the inter-quartile ranges, and the whiskers show the 10th and 90th percentiles, with the median shown as a red line. The y-axis is log-scale. (BISmark)

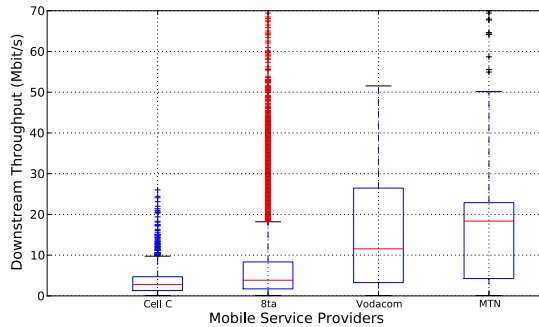


Figure 5. Downstream throughput to a server in Johannesburg, South Africa. Cell C users experience the lowest throughput values; Vodacom and MTN users see the highest throughput. (MyBroadband)

User perceptions. We asked users in the BISmark deployment about their perceptions of speed for common Internet activities. About 75% of our 16 users reported that speeds were acceptable for activities such as Facebook, Instant Messaging (IM) chat, email, and reading news sites, but more than half reported that speeds were too slow for YouTube and video chat. For music downloads, BitTorrent, photo uploads, and watching movies, we received mixed reactions, with many users reporting they do not do these activities. Many users were not satisfied with current speeds for high-bandwidth activities.

4.1.2 How fast are mobile networks?

Using the MyBroadband data, we compared the upload and download speeds for mobile operators. Among our users, MTN and Vodacom users in our study have the highest downstream throughput among these providers, as shown in Figure 5. Performance across different ISPs is variable, particularly for MTN and Vodacom, where users can sometimes experience significantly higher throughputs. The 8ta users also occasionally see high throughputs: throughput measurements above the 90th percentile (shown as plus signs on the graph) are high.

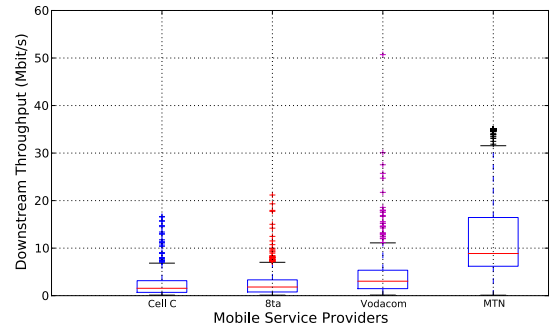


Figure 6. Downstream throughput to a server in London, UK. The relative speeds of each ISP are the same as for the measured performance to Johannesburg, although median downstream throughputs are slightly less since the servers are further away. (MyBroadband)

These trends are similar for these users when measuring throughput against a server in London; Figure 6 shows that, while throughput on paths from South Africa to London is less than to Johannesburg (likely because TCP throughput is inversely proportional to round-trip latency), MTN and Vodacom still perform better than Cell C and 8ta. Performance is also less variable than it is to Johannesburg, presumably because the larger latency to London introduces a performance bottleneck. We see similar trends in the measurements of upstream throughput.

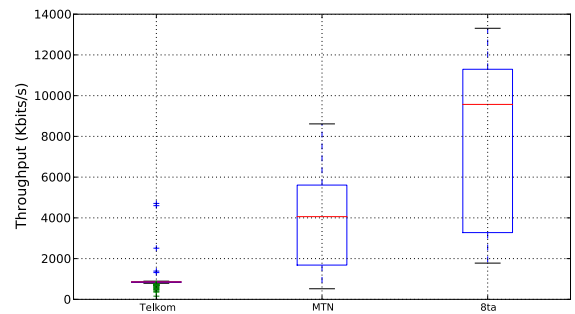


Figure 7. The throughput of one fixed line connection (Telkom) and two mobile connections (MTN and 8ta) from the same location in Cape Town, South Africa to the Measurement Lab server in Johannesburg, South Africa. The fixed-line throughput is considerably lower, but it is also far more consistent (less variable) than either of the two mobile providers. (BISmark)

4.1.3 Which is faster: fixed or mobile?

We now compare the quality of mobile and fixed broadband connections both from a single location using the BISmark data, and using measurements from a larger number of mobile devices and end hosts in the MyBroadband data. Our results are surprising: We found that, in contrast to other regions where

similar studies have been performed (e.g., [23,24]), the performance of fixed-line providers lags significantly behind mobile providers.

Experiment 1: Measurements for both fixed and mobile providers from a single geographic location. We first performed longitudinal measurements from a single location, with both a fixed-line connection (Telkom) and two 3G “dongles” (MTN and 8ta) as explained in Section 3.2. The advertised throughput for the fixed-line connection was 4-10 Mbps, and throughput for the mobile dongles were 5.76 Mbps and 7.2 Mbps, respectively. Figure 7 shows the distribution of throughput values for measurements from this location to a server in Johannesburg, South Africa; the fixed line provider offers consistently lower throughput (even though the advertised speeds of the three connections were comparable). In contrast, the mobile providers offer higher throughput, but the throughput is more variable. Figure 8 shows the same distribution for latencies to the server in Johannesburg; latencies for mobile providers are higher than those to the fixed-line provider from the same location, although all three providers see high variability.

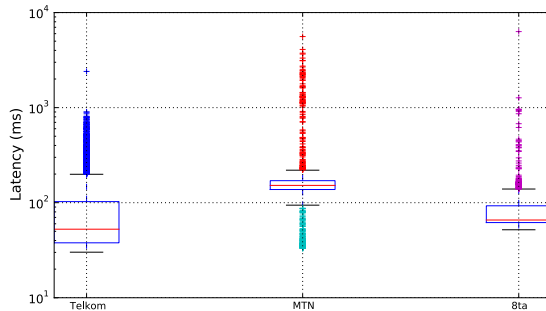


Figure 8. Latency of one fixed line connection (Telkom) and two mobile connections (MTN and 8ta) from the same location in Cape Town, South Africa to the Measurement Lab server in Johannesburg, South Africa. The y-axis is a log scale, so small vertical differences are significant. (BISmark)

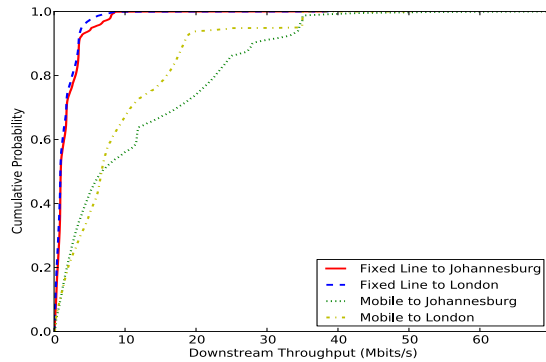


Figure 9. A cumulative distribution function showing the distribution of downstream throughputs for fixed line and mobile users from the MyBroadband data. Mobile throughput is higher than fixed line throughput for both destinations shown. Additionally, throughput is higher to local destinations. (MyBroadband)

Experiment 2: Analyzing speed test measurements from fixed and mobile connections. We also analyzed measurements from the MyBroadband data to compare fixed and mobile connections. Figure 9 shows a distribution of the downstream throughput achieved from MyBroadband measurements for fixed and mobile

users in the country. The plot shows that the median throughput for mobile users is around 10 Mbps, whereas for fixed-line users, the median throughput is closer to about 1 Mbps. Almost none of the fixed-line users achieve performance greater than 1 Mbps. Throughput is also greater to local servers than it is to servers in Europe, for both fixed and mobile. This phenomenon makes sense, given that the throughput of a connection is inversely proportional to the latency of that connection [7]. In Section 4.2, we explore how the latency to various destinations can affect performance to popular sites.

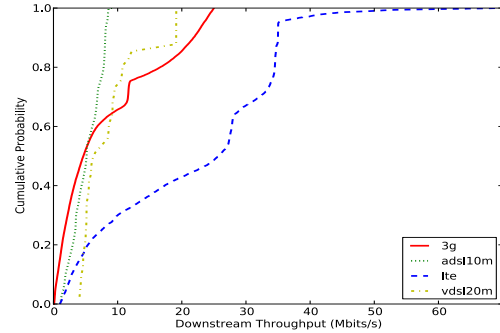


Figure 10. Cumulative distribution of downstream throughput for different fixed-line and mobile network technologies. The mobile technologies (particularly LTE) yield higher throughput than the fixed-line service plans. (MyBroadband)

We also used the MyBroadband data to explore throughput by network type, for both fixed and mobile, to compare the performance of different network technologies. Figure 10 shows the cumulative distribution function of downstream throughput for two different mobile technologies (3G and LTE) and two different fixed-line technologies (ADSL and VDSL). The median downstream throughput of the LTE connections are the highest, at around 25 Mbps. At least half of 3G connections experience higher downstream throughput than the fixed-line service plans. To summarize, both experiments using router-based and host-based measurements in South Africa suggest that mobile broadband is far outperforming fixed line broadband.

4.2 What Other Factors Affect Performance?

We now explore how other factors affect performance, such as the proximity of various servers and services to users; and the time of day (e.g., peak vs. off-peak usage time). We find many popular sites have high-latency paths to and from South Africa. Sometimes, high latencies do not correlate with geographic distance, simply because peering (i.e., the connectivity between different ISPs) is not as dense in Africa as it is in other parts of the world. We discuss these issues in further detail below.

4.2.1 The network paths from servers to users

Latency to servers. Figure 11 shows the average latencies to Google Measurement Lab servers around the world from the BISmark routers deployed across South Africa. The bars are sorted left-to-right in order of increasing distance from Johannesburg. Interestingly, *network latency does not correlate with geographic distance*. For example, network paths to Nairobi have almost twice as high network latencies than the paths to London and Amsterdam, despite the fact that the European destinations are three times further away from Johannesburg, in terms of geographic distance.

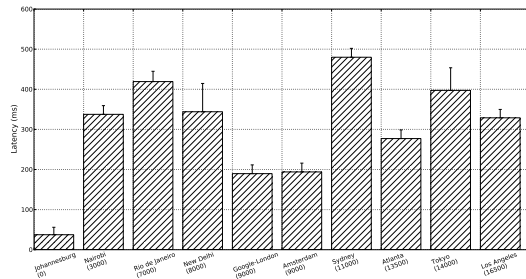


Figure 11. The average latencies to Measurement Lab servers around the world from South Africa. The numbers below each location reflect the distance from Johannesburg in kilometers, and the bars are sorted in order of increasing distance from Johannesburg. (BISmark)

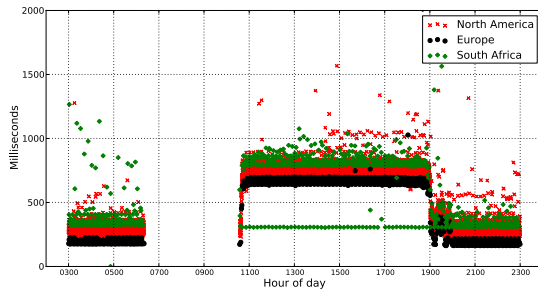


Figure 12. The latencies in milliseconds from BISmark routers around the world to a server in Nairobi, Kenya, on 26 March 2013 when a fiber cut occurred. One South African ISP, Neotel, had better connectivity to Nairobi than other ISPs in South Africa. (BISmark)

We investigated these paths using `traceroute` and found that these inconsistencies arise because the network paths between South Africa and Kenya go through Internet exchange points in Europe, rather than traversing a direct path from South Africa to Kenya. Similarly, latencies to Rio de Janeiro and New Delhi are also relatively high because there are no direct paths between South Africa and either Brazil or India. Although there may now be more direct *fiber* paths between South Africa and India than there were ten years ago [12], fiber alone is not enough to guarantee efficient Internet routing: In order for paths to take more direct routes, ISPs in these respective locations must decide to peer with one another at an intermediate exchange point; often, the only place where both ISPs that peer with one another both connect is an exchange point in Europe.

Latency to popular websites. We measured the latency that users achieved in the country to the Alexa “Top 100” websites using our BISmark. Users experienced average round-trip latencies exceeding 200 ms to five of the ten most popular websites in South Africa: Facebook (246 ms), Yahoo (265 ms), LinkedIn (305 ms), Wikipedia (265 ms), and Amazon (236 ms). Interestingly, our BISmark users perceived the speed for Facebook as acceptable as discussed in Section 4.1.1. However, these latencies suggest that the content for these sites is hosted far away from South Africa; indeed, many of these sites only have data centers in Europe and North America [21]. Previous work has shown that such latencies are high enough to adversely affect user experience [15,33]; these latencies might ultimately be reduced by deploying web caches closer to users or by improving the connectivity between Internet service providers through local exchange points; Section 5 discusses these possibilities.

Reliability to services. The routing from clients in South Africa to global destinations affects not only performance, but also

reliability. When undersea cables were cut off the coast of Egypt on 27 March 2013 [20], we observed an increase in latency to measurement servers from many global destinations for fixed-line users, particularly Nairobi. According to the Renesys blog [20], the cable cut occurred at 06:20 and the server in Nairobi was completely unreachable from *any* location from 06:20:45 to 10:34:38.

Figure 12 shows the timeline of events corresponding to the fiber cut. The figure shows performance measurements to Nairobi not only from BISmark deployments in South Africa, but also from BISmark deployments in other geographic regions, as baselines for comparison. All locations saw interrupted connectivity to Nairobi during the gap in the plot. Interestingly, when the Nairobi server came back online, it experienced significantly higher latencies from all other locations around the world for several hours, except from one ISP in South Africa (AS 36937, Neotel).

This result suggests that Neotel may have better connectivity to destinations within Africa than some other ISPs, and that access ISPs who use Neotel for “transit” may see better performance and reliability to destinations within the continent. Because only the SEACOM cable was affected by the cut, not the West African Cable System (WACS) or EASSy cable [1], Neotel’s access to other fiber paths may have allowed its users to sustain better performance after the fiber cut.

4.2.2 Time of day

Previous studies have shown that the local time of day affects the network performance that different clients experience because the network can be more congested during “peak” hours [31,32]. To determine whether similar phenomena exist in South Africa, we measure whether time of day affects the performance of mobile networks using the MySpeedTest data.

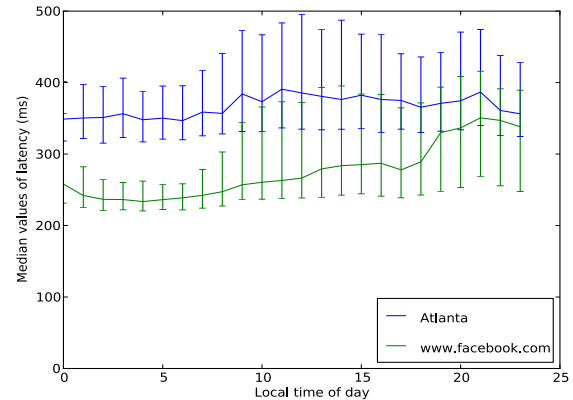


Figure 13. Median latency values by local time of day to Facebook and a server in Atlanta, US). Latencies to both servers are incredibly high (more than 200 ms), and latencies increase and become more variable during peak hours. (MySpeedTest)

We measured how performance varies according to time of day to the US and popular sites such as Google and Facebook. Figure 13 shows the median latency values to a server in Atlanta and to Facebook by hour of the day. These results illustrate that (1) performance to popular sites such as Facebook remains incredibly slow by contrast, most users in the United States and many developed countries see latencies less than 100 milliseconds to sites such as Facebook [6]; (2) performance is worse and more variable during “peak” times—during the day, and at night.

Latency to Facebook from mobile devices in South Africa increases dramatically at around 6 p.m. local time, for example.

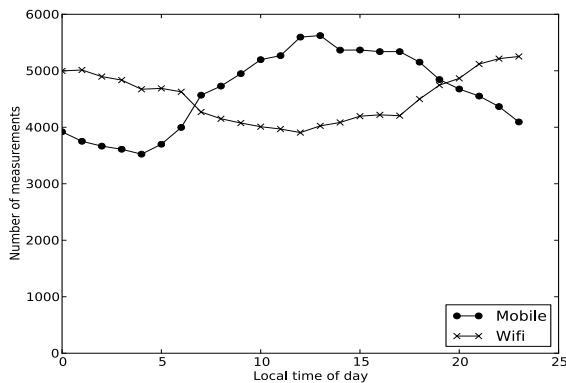


Figure 14. The average number of MySpeedTest measurements per hour, by local time of day, for both mobile and WiFi. From approximately 7 p.m. to 6 a.m. local time, WiFi measurements exceed mobile measurements, suggesting that more users tend to “offload” their traffic to WiFi during times when they are likely to be home. (MySpeedTest)

Relatively more users shift to WiFi networks during the hours when they are likely to be home. Figure 14 clearly demonstrates this behavior; the figure shows that the proportion of users who are on WiFi vs. mobile data networks shifts during evening hours when users are not at work. As in other countries, mobile data networks are significantly less congested during evening and night hours, whereas home WiFi users exhibit the opposite trend.

5. DISCUSSION & RECOMMENDATIONS

We now offer takeaways and recommendations for measuring and improving broadband performance for users in South Africa. We expect that these recommendations may also apply in similar developing countries, particularly in sub-Saharan Africa.

First, our data analysis suggests that users in South Africa are not getting the speeds that ISPs are advertising. Second, our results confirm that mobile broadband is not only more affordable [30], but it is also outperforming fixed broadband in the country. Third, throughput is not the only limiting factor on broadband performance in South Africa. Instead, the high latency to popular websites is also hindering performance. For these reasons, we make two policy recommendations for South Africa.

Institute continual measurement of broadband performance in South Africa for both fixed and mobile broadband providers. The continuous BISmark measurements allowed us to inform certain users that they were not getting advertised speeds. In one extreme case we described in the paper, the user in question has asked ISP to rectify the issue of receiving 1 Mbps on a connection that promises between 4–10 Mbps. On the other hand, host-based measurements allowed us cover the performance of many ISPs more broadly. Using a combination of approaches was most effective for an in-depth and broad analysis of broadband performance on both fixed and mobile networks. Router-based measurements require more upfront initial investment and a deployment on the order of 50–100 across a country to be representative. Host-based measurements can achieve widespread coverage more quickly, since installation typically only requires installing software on devices that users already own or clicking on a web based application. We therefore recommend using a combination of these approaches to ensure

both broad and continuous coverage. We note that simply deploying a technical measurement platform is not sufficient to sustain a measurement effort: an in-depth understanding of each country’s unique conditions is essential for both the sustainability of the infrastructure and the ability to interpret the results. Building partnerships with local organizations for each countrywide measurement study can help facilitate this process. For example, in our ongoing work, we are partnering with local Internet Society chapters to help us identify local partners.

Facilitate investment in local server and service infrastructure to reduce latency. Our results imply that low broadband throughput is often not the only cause of poor performance. In particular, latency to the servers hosting a particular service has a significant effect on the quality of connectivity, particularly to popular websites and services. For this reason, aside from improving the speeds in the country, policymakers could focus on incentives for encouraging private investment in local infrastructure that is closer to South African users or creating an environment that encourages local interconnection among ISPs, both of which would decrease the latency to commonly used services. Reducing the distance of Internet content to South African users will significantly improve the user experience for some of the most popular sites. The following actions are needed:

- *Improve geographic proximity to content with local data centers and hosting.* Corporations that deliver popular (or important) services to users should be encouraged to build data centers and hosting facilities in locations that are closer to users, so that content can be geographically close.
- *Improve Internet paths to content with local Internet exchange points.* The Internet Service Providers’ Association (ISPA) currently operates two Internet exchange points: the Johannesburg Internet exchange (JINX) and the Cape Town Internet Exchange (CINX), yet many local ISPs do not connect at these exchange points because doing so is too costly. To obtain global connectivity, these ISPs must already “backhaul” to major European IXPs (e.g., LINX, AMS-IX), and many have no incentives to pay the extra costs to peer locally. Thus, even if content providers do deploy regional web caches, users of some ISPs may still take circuitous Internet paths to that content if local interconnection arrangements do not exist. Local governments could encourage ISPs in SSA to connect and peer at local or regional exchange points.

6. CONCLUSION

We have presented the results of our ongoing study of fixed and mobile broadband performance in South Africa using an open-source technical platform that we implemented and adapted for a developing country context. Our results suggest that (1) users are not getting the speeds that their ISPs advertise; (2) mobile broadband users commonly achieve higher throughput than fixed-line users, although the performance that they achieve is variable; and (3) speed (*i.e.*, throughput) is not the only limiting factor on performance in South Africa, and that latency to popular websites and services—as determined by both geographic location and Internet peering relationships—may significantly affect performance. Thus, many of the positive network effects promised by broadband connectivity may not be realized in South Africa; rather the widespread problems associated with throughput and latency may ultimately affect the ubiquity of cloud and other services as they extend beyond corporate use.

As such, policymakers should implement broadband performance monitoring using both host-based and router-based approaches. Regulators should also create a favorable environment for private investment in local infrastructure and services that will move Internet content closer to South African users. We expect that our approach and tools can apply in other developing countries, to help inform regulators about how to invest in infrastructure that will improve the performance that users experience. We are expanding our study both within South Africa and to other countries in SSA. The data from our study is available at <http://projectbismark.net/>.

7. ACKNOWLEDGMENTS

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